

Taxi Demand Prediction using Ensemble Model Based on RNNs and XGBOOST

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Abstract — Taxis play an important role in urban transportation. Understanding the taxi demand in the future gives an opportunity to organize the taxi fleet better. It also reduces the waiting time of passengers and cruising time of taxi drivers. Even, there are some works proposed to predict the demand of taxi but there are few studies that consider the function of areas such as hospital area, department store area, residential area, and tourist attraction. One predictive model may not fit with all types of area. We use a point of interest (POI) to match taxi demand with a place to study the taxi demand in the area with a different function. In this paper, we investigate the best predictive models that can forecast demand of taxi hourly with 7 types of area function. The models that were selected for the experiment are long short term memory (LSTM), gated recurrent unit (GRU) and extreme gradient boosting (XGBOOST). Then, we proposed the ensemble model that can forecast the taxi demand well with all types of area function using the information from those machine learning models. We build the models based on a real-world dataset generated by over 5,000 taxis in Bangkok, Thailand for 4 months. The result shows that the proposed ensemble model can outperform other models in overall.

Keywords— Taxi Demand Prediction; Time Series; Neuron Networks; Spatio-temporal Data

I. INTRODUCTION

Where to find a passenger is one of the most important questions for all taxi drivers. The more time taxi driver spends on cruising a new passenger, the more fuel consumption and the fewer number of passengers can be picked-up. For inexperienced taxi drivers, they usually don't know where to pick-up a new passenger since they have no experience about the demand of taxi over time and space. The information about taxi demand in the future can be used to guide both inexperienced and experienced drivers to catch up with the taxi demand in the city faster. Hence, it helps to match the demand with supply in taxi services.

With Rich information from taxi GPS sensors, many researchers have used this digital footprint to discover some knowledge that can help us improving the transportation system.

Therefore, there are many existing proposed systems which use taxi GPS trajectories as their transportation sensors. Anomaly detection is one of the most popular topics using taxi GPS data. Zhang, D. et al [1], applied isolation forest method which uses “Few and Different” concept to develop anomaly detection system in taxi trajectories. Yisheng Lv. et al [2], used a GPS trace to build a deep learning model to predict traffic flow of a city.

For taxi demand prediction studies, Moreira-Matias, L., et al [3], presented a model for predicting the number of services that will happen at taxi stands by applying the time-varying Poisson model and the auto regression integrated moving average (ARIMA). Moreover, they used sliding-window ensemble framework to originate a prediction by combining the prediction of each model. The dataset was generated from 441 vehicles with 63 taxi stands in the city of Porto. N. Davis et al [4], proposed a taxi demand forecasting model using multi-level clustering approach and data from a leading taxi booking application in the city of Bengaluru, India. They applied many linear forecasting models such as Holt-Winters (HW) model, Seasonal Naive, STL decomposition, ARIMA, TBATS. The result showed that STL had the best performance. There are also studies [5], [6], [7] using artificial neural networks to forecast demand of taxi. Mukai and Yoden [5], also applied artificial neural networks to predict the taxi demand in 25 regions of Tokyo. They divided the hours of a day into 6 intervals (4 hours). They reported that the area that has highest error rate is the area that has many types of transportation such as railway, subway because the taxi demand is so small and non-periodical. Zhao et al [6], compared Markov predictor with Neural Network predictor using the dataset from yellow taxicab and Uber taxi in New York. The result showed that Markov predictor can achieve high accuracy in the area with high theoretical maximum predictability while neural network model performed better in the area that has lower theoretical maximum predictability. Jun Xu et al [7], divided the entire New York City into around 6,500 areas. Then, they applied a special kind of recurrent neural network which is long short term memory recurrent neural

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network (LSTM RNN) with a data sequence of the number of taxi requests in each area.

In this paper, we proposed an ensemble model to predict the demand of taxi using three machine learning models which are long short term memory (LSTM) [8], gated recurrent unit (GRU) [9] and extreme gradient boosting (XGBOOST) [10]. Unlike previous works, we aggregated demand of taxi within 200 meter radius of a point of interest (POI) which is a place such as a hospital, an airport, and a subway. We investigated how well each model can perform in the areas with a different function. Then, we combined the result produced by those models to produce a final prediction. Figure 1 shows taxi demand patterns of areas with a different function.

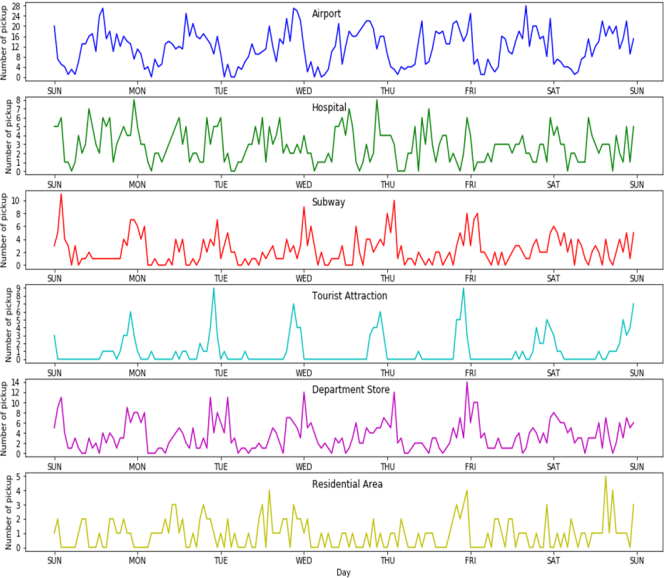


Fig. 1. Demand patterns of area with different function

The rest of the paper is organized as follows. In Section II, we describe about data processing. In section III, we briefly explain about the chosen predictive models. In section IV, we show the experimental setup and result. Lastly, we conclude everything in the section V conclusion.

II. DATA PROCESSING

The dataset using in this study is real-world data generated by taxis GPS in Bangkok, Thailand from January 1 to June 30, 2016. The data collection process was supported by Toyota Tsusho Electronic (Thailand) Co, Ltd. The sampling rate of this dataset is 5 second. For historical weather data in Bangkok, we use the data from openweathermap.org which has hourly sampling rate. There are many types of weather condition obtained from the website such as Thunderstorm, Rain, Clouds, Clear, and Frog. We reclassify the weather data into only clear and rainy. The detail of each fields in datasets is shown in table I.

TABLE I. SPECIFICATION OF TAXI TRACE

Data Type	Description
IMEI	The taxi unique ID

Timestamp	Timestamp of the sample point
Latitude/Longitude	GPS location of the sample point
Speed	Current speed of a taxi at the sample point
Angle	Current direction of a taxi at the sample point
Meter	The status which indicates whether a taxi is occupied or not (the status is 0 when available, and become 1 when occupied)

A. Data Preparation

The GPS sensor in a taxicab is very sensitive. Therefore, the obtained position sometimes may be invalid. We want to create a reliable model to predict the taxi demand accurately, thus the invalid data should be removed. There are four steps in this process.

1) Extracting pick-up/drop-off from raw GPS record

A shift of meter status indicates a pick-up/drop-off event. If the meter is changed from 0 to 1, we mark that position as the pick-up point. On the other hand, if the meter is changed from 1 to 0, the position is marked as the drop-off point.

2) Filtering invalid trajectories

After preliminary investigating the average duration and speed of normal trajectories, the normal range of duration should be 100 to 6,000 second. The normal range of speed should be between 1 m/s to 30 m/s. The trajectories which are out of those ranges will be removed.

3) Mapping the pick-up event to POI

In this study, we use Points of Interest (POI) to match pick-up event with an area. If the pick-up event is within a radius of 200 meters from a point of interest, we will consider that the pick-up event occurred at that POI.

4) Calculating the aggregated demand in every hour for each point of interest

The number of pick-up events represents the demand of taxi. We aggregated the number of pick-up events for all areas in every hour. Therefore, we have the taxi demand of each area in 24-time slots a day.

B. Point of Interest

There are 7 types of area function in this study. We select top 10 taxi demand places for each type except the airport because Bangkok has only 2 airports.

- 1) Hospital
- 2) Residential area
- 3) Education area
- 4) Tourist attraction area
- 5) Airport
- 6) Department Store
- 7) Subway

C. Features

There are 6 main features for the predictive model. These features are selected based on previous studies [5], [7].

- 1) Number of pick-up
- 2) Hour of a day
- 3) Day of week
- 4) Day of month
- 5) Weather
- 6) National Holiday

We also applied rolling window technique with the number of pickups feature. Let $n(t)$ be the number of pick-up in time step t and w be the window size that we want to look back, if we want to predict $n(t)$, we will also use $n(t-1)$, $n(t-2)$, ..., $n(t-w)$ as extra features. The window size w is 24 in this study.

III. MODELS

In this section, we briefly explain about our selected models which are Long Short Term Memory (LSTM), Gated Recurrent Unit (GRU) and eXtreme Gradient Boosting (XGBOOST). We also describe the overview of ensemble model in this section.

A. Recurrent Neural Network (RNN)

A recurrent neural network is a special kind of artificial neural network which has a backward connection between hidden layers. The recurrent neural networks have internal memory which allows them to operate over sequential data effectively. Therefore, RNNs are one of the most popular models for dealing with sequential tasks such as handwriting recognition, language modeling and time series prediction.

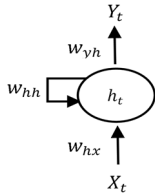


Fig. 2. A recurrent neural network

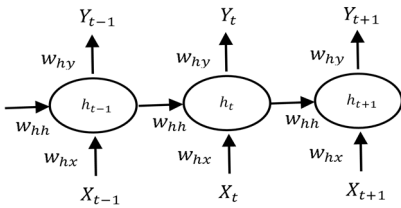


Fig. 3. An unrolled recurrent neural network

Figure 2 shows the structure of a recurrent neural network. Figure 3 represents the unrolled version of a recurrent neural network which illustrates how the network deal with sequential data clearly. Given an input of time series $X = \{x_1, x_2, \dots, x_t\}$, the RNN computes the hidden state sequence $H = \{h_1, h_2, \dots, h_t\}$ and output sequence $Y = \{y_1, y_2, \dots, y_t\}$ iteratively using (1), (2).

$$h_t = f(W_{hx}x_t + W_{hh}h_{t-1} + b_h) \quad (1)$$

$$y_t = g(W_{yh}h_t + b_y) \quad (2)$$

In (1) and (2), W_{hx} , W_{hh} and W_{yh} denote the input-to-hidden weight matrix, the hidden-to-hidden weight matrix and the hidden-to-output weight matrix respectively. The vector b_h and b_y are the bias of the hidden layer and the output layer respectively. $f(\cdot)$ and $g(\cdot)$ are the activation functions of the hidden layer and output layer respectively. The hidden state of time step t will be passed to the hidden state of time step $t + 1$.

However, the conventional recurrent networks suffer from the gradient vanishing problem when it works with multi-step dependencies. Long short term memory networks [8] were invented by Hochreiter and Schmidhuber in 1997. It was explicitly designed to avoid the gradient vanishing problem in the traditional recurrent neural networks.

1) Long short term memory

Long short term memory or LSTM is one of the most popular artificial neural networks currently because it can achieve high accuracy result in many kinds of research. The repeating module in LSTM has more complicated structure than the traditional RNNs. It contains a cell state, a forget gate, an input a gate and an output gate.

2) Gated recurrent unit

Gated recurrent unit was introduced by Cho et al. [9] in 2014. The structure of gated recurrent unit is similar to LSTM. GRU cell consists of two gates which are a reset gate and an update gate. In some researches, GRU provides a comparable result to LSTM but for taxi demand prediction, it hasn't been explored yet. Therefore, we selected GRU to be another predictive model in this study.

B. Extreme gradient boosting

Gradient boosting tree model was proposed by Friedman et al in 2001 [11]. The main idea of boosting is to merge a set of weak learner to a strong one, in an iterative fashion. Extreme gradient boosting (XGBoost) [10] is a library optimized for boosting algorithm. The library provides a scalable, portable framework which was used in many data science competitions.

C. Ensemble model

We combine the predictions from those three models to find the best final prediction by assigning more weight to the model that achieved low error rate in the previous w time steps. We use sMAPE (Symmetric mean absolute percentage error) to evaluate the performance of each model. The formula of sMAPE is described in section IV. Firstly, we find the average of sMAPE in previous w time steps for all models. Let $E_m = \{e_{m(t-w)}, \dots, e_{m(t-3)}, e_{m(t-2)}, e_{m(t-1)}\}$ be the set of sMAPE values for model m in the previous w time steps. We calculate the average sMAPE of model m in the previous w time steps using (3).

$$E_{m(avg)} = \frac{1}{w} \sum_{i=1}^w e_{m(t-i)} \quad (3)$$

Then, we rank those models based on their average sMAPE in the previous w time steps. Let P_f be the final prediction and $P_{m^{1st}}, P_{m^{2nd}}, P_{m^{3rd}}$ be the predictions from the models those were ranked in ascending order. μ, β, α are the coefficients of the first rank model, the second rank model and the third rank model respectively. Finally, the final prediction P_f can be computed by the weighted arithmetic mean in (4). Figure 4 illustrates the architecture of our ensemble model.

$$P_f = \mu P_{m^{1st}} + \beta P_{m^{2nd}} + \alpha P_{m^{3rd}} \quad (4)$$

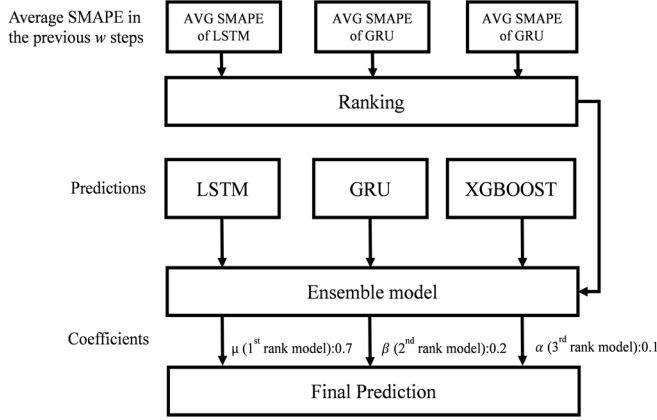


Fig. 4. Architecture of our ensemble model

IV. EXPERIMENTAL RESULTS

This section contains the experimental setup (A), the evaluation metric (B), and the experimental result (C).

A. Experimental setup

We have a real-world dataset generated from over 5,000 taxis in Bangkok for 4 months. We train our models with the first 3 months and validate those models with the last month. The long short term memory and gated recurrent unit models were implemented in Keras API [12] which is built on top of Tensorflow. The simple architecture achieves low error rate. The designed architectures for both LSTM and GRU have one hidden layer which contains 150 neurons. The library [10] is used to build the gradient boosting tree model. For the ensemble model's coefficients, μ, β, α are 0.7, 0.2, and 0.1 respectively. The window size w is 3.

B. Evaluation Metric

We evaluate the performance of our predictive models with the widely used prediction error metrics: Symmetric Mean Absolute Error (sMAPE). The formulation of the prediction error metric is given as (5).

$$sMAPE_i = \frac{1}{T} \sum_{t=1}^T \frac{|Y_{i,t} - \hat{Y}_{i,t}|}{Y_{i,t} + \hat{Y}_{i,t} + C} \quad (5)$$

$Y_{i,t}$ is the real demand while $\hat{Y}_{i,t}$ is the predicted demand occurred in area i at time t . We also include constant C which equals one to prevent division by zero.

C. Experimental result

We show the sMAPE results of all models over all types of area function. Then, we compare the predictions of all models with the real demand on a specific place.

1) sMAPE results over all types of area function

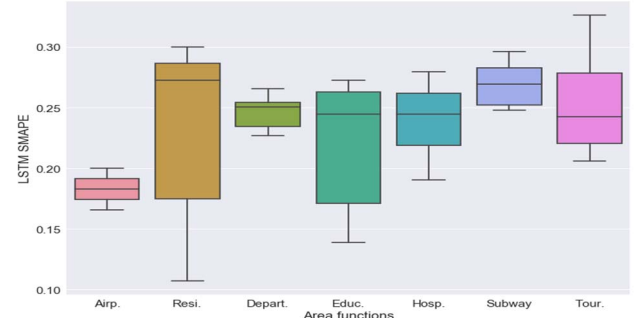


Fig. 5. sMAPE result from the LSTM model

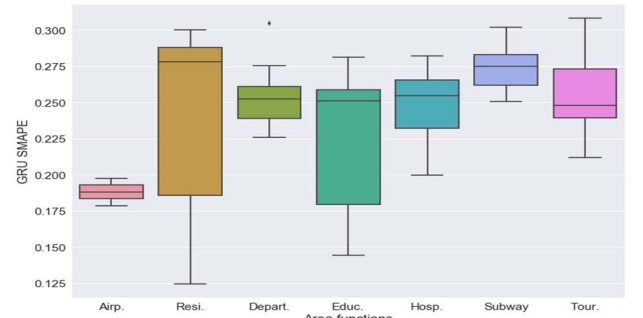


Fig. 6. sMAPE result from the GRU model

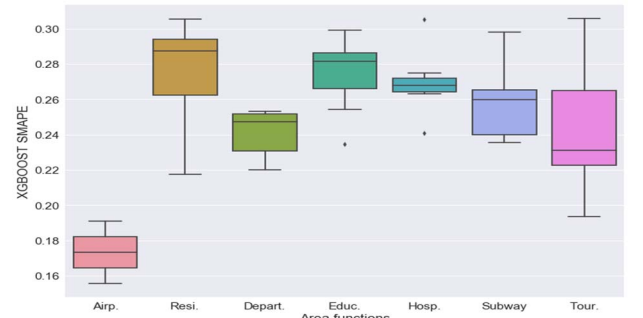


Fig. 7. sMAPE result from the XGBOOST model

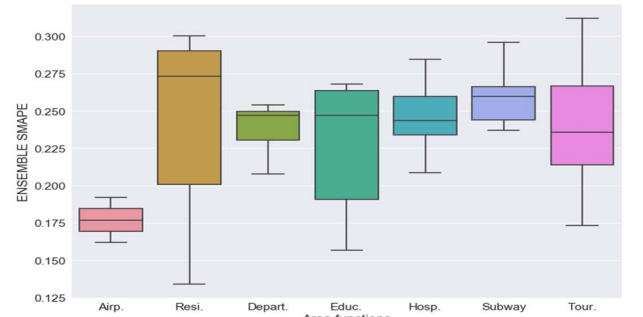


Fig. 8. sMAPE result from the ensemble model

Figures 5,6,7,8 show the sMAPE result of the LSTM, GRU, XGBOOST and ensemble models in box plot respectively. The area that all models achieve the lowest sMAPE result is the airport area.

TABLE II. AVERAGE SMAPE OF ALL MODELS IN PERCENT

Areas	Models			
	<i>lstm</i>	<i>gru</i>	<i>xgboost</i>	<i>ensemble</i>
Airport	18.30%	18.80%	17.31%	17.62%
Residential Area	23.32%	23.95%	27.42%	24.01%
Department Store	24.65%	25.40%	24.21%	23.95%
Education Area	22.01%	22.23%	27.41%	22.20%
Hospital	24.01%	24.84%	26.92%	24.52%
Subway	27.01%	27.44%	25.75%	25.92%
Tourist Attraction	25.39%	25.44%	24.10%	23.85%
All places	24.32%	24.82%	25.55%	24.02%

The average sMAPE of all models in each area types were shown in table II. The XGBOOST approach can outperform both LSTM and GRU model in the airport, department store and tourist attraction areas which are high taxi demand areas. The prediction of GRU is really closed to LSTM’s but the result shows that LSTM still provides better prediction than GRU for all types of area. Both LSTM and GRU work very well in low taxi demand areas such as residential area, education area and hospital area. With sMAPE of 24.32%, LSTM is the best independent predictive model for taxi demand prediction in this experiment. For the ensemble model, even though, it can provide the lowest error rate in only tourist attraction and department store areas but it produced the results that are closed to the result of the best model in each type of area function. The average sMAPE of the ensemble model in all places is 24.02% which is the lower than all standalone models.

2) Comparing the predictions over a specific place

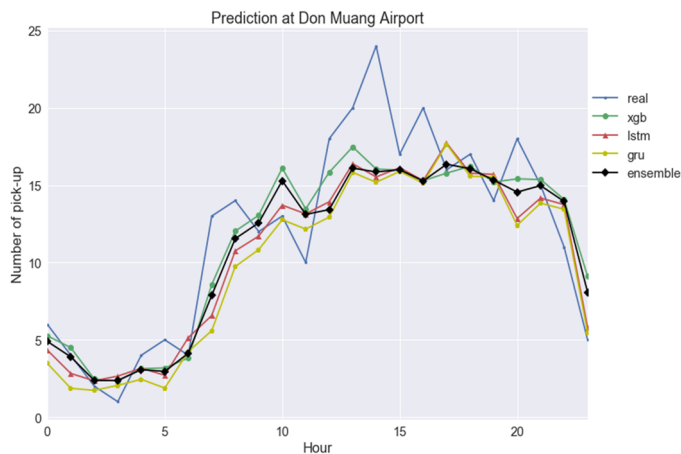


Fig. 9. One day prediction in the airport

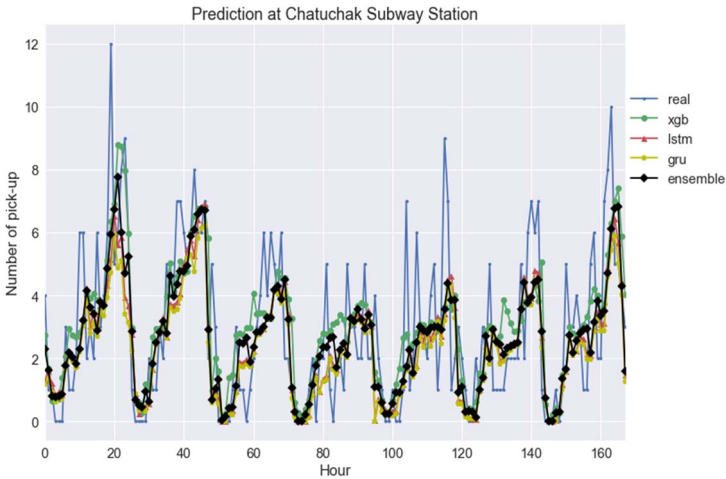


Fig. 10. One week prediction in the subway area

Figure 9 shows the comparison of one day prediction of each model with the actual demand in the airport while figure 10 shows the comparison of one week prediction in the subway area. Those figures 9, 10 illustrate that the model which provides a better prediction lately is likely to produce more accurate prediction in the next time step. The ensemble model follows this trend. Therefore, it can provide a better prediction in overall.

V. CONCLUSTION

We proposed an ensemble model based on long short term memory network (LSTM), gated recurrent unit network (GRU) and eXtreme gradient boosting (XGBOOST) models to predict taxi demand in 7 types of area function in the city of Bangkok, Thailand. We use Point of Interests (POIs) to match the taxi demand with studied areas. The model that provides a better prediction lately is likely to produce more accurate prediction in the next time step. The ensemble model predicts taxi demand by following this trend. The results show that the ensemble model outperforms other standalone models with sMAPE of 24.02% in all areas. For the independent models, LSTM provides the most accurate prediction in low taxi demand areas such as residential area, hospital and education area. LSTM achieves better results than GRU in all types of area. The XGBOOST model gives better results than the others in high taxi demand areas such as department store, subway and airport areas. This study shows that a single predictive model cannot provide the best prediction in all areas. Combining the predictions from various models can improve the performance in overall.

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REFERENCES

- [1] ZHANG, D., LI, N., ZHOU, Z.-H., CHEN, C., SUN, L., AND LI, S., "IBAT: detecting anomalous taxi trajectories from GPS traces," In 13th conference on ubiquitous computing, UbiComp, 2011.
- [2] Y. Lv, Y. Duan, "Traffic flow prediction with big data: A deep learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 2, pp. 865-873, 2015.
- [3] Moreira-Matias, L., et al., "Predicting Taxi–Passenger Demand Using Streaming Data," *IEEE Transactions on Intelligent Transportation Systems*, vol. 14, no. 3, pp. 1393-1402, 2013.
- [4] N. Davis, G. Raina, and K. Jagannathan, "A multi-level clustering approach for forecasting taxi travel demand," in *Proc. IEEE ITSC*, Dec, pp. 223-228, 2016.
- [5] Naoto Mukai and Naoto Yoden, "Taxi Demand Forecasting Based on Taxi Probe Data by Neural Network," *Smart Innovation, Systems and Technologies*, 2012.
- [6] Kai Zhao, Denis Khryashche, Juliana Freire, Cláudio Silva, and Huy, "Predicting Taxi Demand at High Spatial Resolution: Approaching the Limit of Predictability," in *IEEE International Conference on Big Data*, 2016.
- [7] Jun Xu et al, "Jun Xu et al, "Real-Time Prediction of Taxi Demand Using Recurrent Neural Networks," *IEEE Trans. Intell. Transp. Syst.*, vol. pp, no. 99, pp. 1-10, 2017.
- [8] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [9] K. Cho, B. van Merriënboer, C. Gulcehre, F. Bougares, H. Schwenk, D. Bahdanau, Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation", *arXiv preprint arXiv: 1406.1078*, 2014.
- [10] Tianqi Chen, Carlos Guestrin "XGBoost: A Scalable Tree Boosting System", *arXiv: 1603.02754v3 [cs.LG]* 10 Jun 2016.
- [11] Friedman, Jerome H. "Greedy function approximation: a gradient boosting machine." *Annals of statistics* (2001): 1189-1232.
- [12] Chollet, François, Keras, (2015), GitHub repository, <https://github.com/keras-team/keras>